

Course Objectives:

1. Understand the fundamentals of computer science and programming
2. Develop an understanding of the principles of software engineering
3. Develop an understanding of the principles of operating systems
4. Learn how to design and develop algorithms
5. Develop an understanding of data structures and their applications
6. Learn how to design and develop efficient and secure software
7. Understand the principles of computer networks and their applications
8. Develop an understanding of the principles of database systems
9. Learn how to develop software applications for various platforms
10. Develop an understanding of the principles of artificial intelligence and machine learning

1. To enable the students to understand about the Chemistry of Atomic and Molecular structure, Chemistry of

- Atoms are the basic building blocks of matter and are composed of protons, neutrons, and electrons.
- The number of protons in an atom defines its atomic number and determines its chemical element.
- Electrons are found in different energy levels around the nucleus, and the arrangement of electrons in an atom determines its chemical properties.
- Molecules are two or more atoms held together by chemical bonds.
- Chemical bonds are formed when atoms share or transfer electrons.
- The chemical properties of a molecule are determined by the arrangement of its atoms and the types of bonds between them.

Advanced Materials

- Liquid Crystals: Liquid crystals are a state of matter that lies between solid and liquid phases. They are composed of molecules that are arranged in an ordered, but not completely rigid, arrangement. They are characterized by anisotropic physical properties, including optical, electrical, and thermal properties.
- Nanomaterials: Nanomaterials are materials that are composed of particles that are smaller than 100 nanometers in size. They have unique physical, chemical, and biological properties that differ from larger materials.
- Graphite & Fullerenes: Graphite is a form of carbon that is composed of layers of hexagonal lattices. Fullerenes are a form of carbon that is composed of spherical or cylindrical molecules. Both of these materials have unique properties, such as high electrical and thermal conductivity.
- Green Chemistry: Green chemistry is a set of principles that are designed

to reduce or eliminate the use of hazardous materials and processes in the production of chemicals. It is based on the principles of reducing waste, increasing efficiency, and preventing pollution.

2. To enable the students to understand and apply the detailed concepts of spectroscopic techniques and

1. Spectroscopic techniques are used to measure the interaction of electromagnetic radiation with matter.
2. Spectroscopy is used to identify different elements and molecules in a sample, as well as to measure the concentration of the sample.
3. The different types of spectroscopic techniques include absorption spectroscopy, emission spectroscopy, and Raman spectroscopy.
4. Absorption spectroscopy measures the amount of light that is absorbed by a sample.
5. Emission spectroscopy measures the amount of light that is emitted from a sample.
6. Raman spectroscopy measures the amount of light that is scattered by a sample.
7. The different types of spectroscopic techniques can be used to measure the composition, structure, and properties of a sample.
8. The different types of spectroscopic techniques can also be used to identify unknown compounds and to monitor chemical reactions.

Stereochemistry

Stereochemistry is the branch of chemistry that studies the spatial arrangement of atoms within molecules. It is an important part of organic chemistry and is used to identify compounds, elements, and other molecules.

Properties of Stereochemistry

1. Stereochemistry is based on the three-dimensional arrangement of atoms within a molecule.
2. Stereochemistry is used to classify compounds based on their molecular structure.
3. Stereoisomers are molecules with the same molecular formula but different arrangements of atoms in space.
4. Chiral molecules are those that are not superimposable on their mirror image.
5. Chirality is determined by the presence of a chiral center, which is an atom with four different substituents attached.
6. Stereochemistry can be used to determine the absolute configuration of a molecule.
7. Stereochemistry is also used to predict the reactivity and reactivity of a molecule.

3. To enable the students to understand and apply the concepts related to Electrochemistry, Batteries, Corrosion

1. Electrochemistry:
 - Electrochemistry is the study of the relationship between electrical energy and chemical energy. It involves the study of the production of electricity from chemical reactions and the use of electrical energy to cause chemical reactions.
 - Electrochemistry is used in the study of batteries, fuel cells, electrolysis, corrosion, and the production of electrical energy from chemical sources.
2. Batteries:
 - Batteries are devices that store electrical energy and convert it into chemical energy.
 - Batteries are composed of two or more electrochemical cells connected in series or parallel.
 - The chemical reactions in the cells produce electrical energy, which can be used to power electronic devices.
3. Corrosion:
 - Corrosion is a process in which a material is attacked by chemical reactions with its environment.
 - Corrosion is caused by the presence of oxygen, water, and other chemicals that react with the material.
 - The corrosion process can be slowed down by using protective coatings or other methods.

Chemistry of Engineering Materials

- Cement is a type of engineering material made from a mixture of calcium, silicon, iron, and aluminum.
- Cement is a strong and durable material that is used in construction projects all over the world.
- Cement is a versatile material that can be used to make concrete, mortar, and other types of building materials.
- Cement is made by heating limestone and other materials in a kiln at high temperatures.
- The chemical reaction between the ingredients creates a hard material that binds other materials together.
- Cement is used in various applications, including as a binding agent in concrete, as a waterproofing agent in mortar, and as a filler in asphalt.
- Cement is also used in the production of bricks, tiles, and other building materials.
- The properties of cement can be influenced by the type of raw materials used and the manufacturing process.
- Cement can be modified to improve its strength, durability, and other

properties.

4. To Enable Students to Understand and Apply Detailed Concepts of Water Source, Water Impurities, and Hardness

1. **Water Source:** Water sources can be classified as surface water, such as lakes and rivers, or ground water, such as aquifers. Surface water is more exposed to contamination from pollutants, while ground water is more protected, but can still contain pollutants.
2. **Water Impurities:** Water impurities can be classified as organic or inorganic. Organic impurities include bacteria, viruses, and other microorganisms, while inorganic impurities include minerals, metals, and other chemicals.
3. **Hardness:** Hardness is a measure of the amount of calcium and magnesium in the water. Hard water can cause scale buildup in pipes and appliances, and can also affect the taste of the water.

Water and Boiler Troubles Used in Industry

1. Boiler problems can arise due to a number of reasons, including:
 - Poor maintenance
 - Improper installation
 - Faulty equipment
 - Contaminated water
2. Boiler problems can be divided into two categories:
 - Primary problems, which are caused by the boiler itself
 - Secondary problems, which are caused by external factors
3. Common boiler problems include:
 - Leaks
 - Corrosion
 - Boiler overpressure
 - Low water level
 - Sludge buildup
4. To prevent boiler problems, it is important to carry out regular maintenance and inspections.

Analysis of Coal and Determination of Calorific Values

1. Calorific value is the amount of heat released by a unit of coal when it is burned.
2. Calorific values can be determined by analyzing the coal's chemical composition.

3. The analysis of coal involves testing the coal's moisture content, ash content, volatile matter content, and sulfur content.
4. The results of the analysis are used to calculate the calorific value of the coal.
5. The calorific value is then used to determine the efficiency of the coal in producing heat.

5. To enable the students to understand detailed concepts related to polymers, Polymerization, Polymer Blends

1. Polymers are materials composed of long chains of molecules, which are formed when many small molecules (monomers) are joined together.
2. Polymerization is the process of forming polymers from monomers. This can be done through a variety of methods, including chemical reactions, physical processes, and biological processes.
3. Polymer blends are mixtures of two or more polymers that have been combined to create a new material with unique properties. These blends can be used for a variety of applications, including insulation, packaging, and engineering materials.
4. The properties of polymer blends are determined by the type of polymers used, the ratio of the polymers, and the processing conditions.
5. Polymerization can be used to create a variety of polymers, including thermoplastics, thermosets, elastomers, and biopolymers.
6. The physical and chemical properties of polymers can be modified through the use of additives, fillers, and reinforcements.
7. Polymer blends can also be used to create materials with unique properties, such as improved strength, elasticity, and thermal and electrical conductivity.

Polymer Composites

Polymer composites are materials that consist of two or more components, at least one of which is a polymer. These materials are used in a wide range of applications, from automotive and aerospace components to biomedical implants.

Polymer composites can be divided into two main categories: thermoplastics and thermosets. Thermoplastics are polymers that can be repeatedly softened and hardened by changes in temperature, while thermosets are polymers that become permanently hardened when exposed to heat.

The components of a polymer composite can be divided into three main categories: the matrix, the reinforcement, and the interface. The matrix is the main component of the composite and provides the bulk of the material's properties. The reinforcement provides additional strength and stiffness, while the interface helps to bond the matrix and the reinforcement together.

The properties of a polymer composite depend on the type of matrix and re-

inforcement used, as well as the ratio of the two components. For example, a composite made with a stiff matrix and a flexible reinforcement will be more flexible than one made with a flexible matrix and a stiff reinforcement.

Polymer composites are used in a variety of applications, from automotive and aerospace components to biomedical implants. They offer a number of advantages over traditional materials, such as low weight, high strength, and good corrosion resistance.

Content Contact

1. Content contact is the process of engaging with customers through the content they create and consume. It involves using a variety of methods to reach out to customers, such as social media posts, blog posts, email campaigns, and more.
2. Content contact can be used to build relationships with customers, increase brand awareness, and convert leads into customers. It is an important part of any digital marketing strategy.
3. Content contact should be tailored to the target audience. For example, a B2B company might focus on LinkedIn and Twitter, while a B2C company might focus on Instagram and Facebook.
4. Content contact should also be personalized to the customer. It should be relevant to their interests and needs, and should be tailored to their individual preferences.
5. Content contact should be used in conjunction with other digital marketing strategies, such as search engine optimization and pay-per-click advertising. This will help to maximize the reach and effectiveness of the content contact strategy.

Hours

- Hours are a unit of time that is used to measure the duration of an event or activity.
- The hour is the base unit of time in the International System of Units (SI).
- The hour is typically divided into 60 minutes, each of which is divided into 60 seconds.
- The symbol for hour is “h” and it is abbreviated as hr.
- A day is equal to 24 hours.
- A week is equal to 168 hours.
- A month is equal to 730 hours.
- A year is equal to 8760 hours.

Unit-1: 8

1. The term “formal language” refers to a style of writing and speaking that is used in academic, professional, or legal contexts. This type of language is characterized by its precision, clarity, and lack of emotion.
2. Formal language is often used in documents such as contracts, laws, and official documents. It is also used in speeches and presentations to give them a more professional tone.
3. Formal language should be used when writing business emails, memos, reports, and other documents. It should also be used when speaking in a professional setting, such as a job interview or a business meeting.
4. Formal language should be used to convey information in a clear and concise manner. It should avoid using slang, jargon, and colloquialisms.
5. Formal language should be used when writing for an audience that is unfamiliar with the topic. For example, when writing an academic paper, it is important to use formal language that is appropriate for the audience.
6. Formal language should be used when writing for a specific purpose. For example, when writing a legal document, it is important to use language that is precise and accurate.
7. Formal language should be used when communicating with someone in a position of authority. It is important to show respect and professionalism when communicating with someone in a higher position.
8. Formal language should be used when writing for a formal occasion. For example, when writing a wedding invitation, it is important to use a formal tone.

Atomic and Molecular Structure: Molecular orbital's of diatomic molecules, Bond Order

1. Molecular orbitals are mathematical functions that describe the properties of electrons in a molecule.
2. Molecular orbitals are derived from the combination of atomic orbitals from each atom in the molecule.
3. Molecular orbitals are classified as either bonding, antibonding, or non-bonding orbitals.
4. Bond order is a measure of the strength of a chemical bond between two atoms.
5. Bond order is determined by the number of electrons in a bonding molecular orbital minus the number of electrons in an antibonding molecular orbital.
6. The bond order of a diatomic molecule can range from 0 to 4, with 0 indicating that the molecule is not bound and 4 indicating a very strong bond.

Magnetic characters and numerical problems

1. Magnetic characters are symbols that represent the direction and magnitude of a magnetic field. They are typically written as arrows, with the length of the arrow representing the magnitude of the field and the direction of the arrow representing the direction of the field.
2. Numerical problems involving magnetic fields can be solved using the right-hand rule. This rule states that if the right thumb is pointed in the direction of the magnetic field, the fingers will curl in the direction of the force that the field exerts on a moving charge.
3. The force exerted by a magnetic field on a moving charge can be calculated using the equation $F = qv \times B$, where q is the charge of the particle, v is the velocity of the particle, and B is the magnetic field.
4. The Lorentz force law states that the force exerted by a magnetic field on a charged particle is equal to the product of the particle's charge and the cross product of its velocity and the magnetic field. This law can be used to calculate the force exerted by a magnetic field on a charged particle.

Chemistry of Advanced Materials:

- Advanced materials are materials that have been engineered to possess superior properties and performance compared to traditional materials.
- These materials are typically synthesized using chemical techniques, such as polymerization, sol-gel processing, and chemical vapor deposition.
- Depending on the desired properties of the material, different chemical processes can be used to create the desired structure and composition.
- Advanced materials have a wide range of applications, including aerospace and automotive engineering, medical devices, nanotechnology, and energy storage.
- Advanced materials can be divided into three main categories: ceramics, polymers, and composites.
- Ceramics are inorganic materials that are composed of metal oxides, such as alumina and silica.
- Polymers are organic materials that are composed of repeating molecular units, such as polyethylene and nylon.
- Composites are materials that are composed of two or more materials, such as carbon fiber reinforced plastics.
- Each of these categories of advanced materials has unique properties that can be tailored to meet specific design requirements.
- For example, ceramics are often used for high temperature applications due to their thermal and chemical stability, while polymers are often used for lightweight and flexible applications.
- Composites are often used for high strength and stiffness applications due to their ability to combine the properties of two or more materials.

Liquid Crystals; Introduction, Types and Applications of liquid crystals, Industrially

- Liquid crystals are a type of matter that have properties of both liquids and solids. They are composed of molecules that are arranged in an ordered pattern, but are able to move around freely like a liquid.
- There are three main types of liquid crystals: nematic, smectic, and cholesteric. Nematic liquid crystals are the most common and are used in LCD displays. Smectic liquid crystals are used in ferroelectric LCDs, while cholesteric liquid crystals are used in reflective LCDs.
- Liquid crystals have many industrial applications. They are used in LCD displays for computers, televisions, and other electronic devices. They are also used in optical components, such as lenses and polarizers, and in thermometers, thermochromic ink, and other temperature-sensitive materials.
- Liquid crystals can also be used to create smart materials that can react to their environment. For example, they can be used to create fabrics that can change color in response to heat, or materials that can increase their stiffness when exposed to a certain amount of force.

Important Materials Used as Liquid Crystals

1. Calamitic Liquid Crystals: These are the most commonly used liquid crystals, and are composed of rod-like molecules that are arranged in a helical structure. They are typically used in displays, such as those found in televisions and mobile phones.
2. Discotic Liquid Crystals: These liquid crystals are composed of disk-shaped molecules which are arranged in a columnar structure. They are generally used in optical devices, such as lenses and prisms.
3. Thermotropic Liquid Crystals: These liquid crystals are composed of molecules that are arranged in a lamellar structure. They are typically used in thermometers and temperature-sensitive switches.
4. Lyotropic Liquid Crystals: These liquid crystals are composed of molecules that are arranged in a micellar structure. They are usually used in detergents and surfactants.

Graphite and Fullerene; Introduction, Structure and Applications

- Graphite is a naturally occurring form of crystalline carbon. It is composed of layers of carbon atoms arranged in a hexagonal lattice, which gives it its unique properties. Graphite is a very soft material and is often used in pencils, lubricants and as a conductor in electrical components.

- Fullerene is a form of carbon consisting of spherical molecules made up of 60 or more carbon atoms. It has many unique properties, such as being the strongest material known and having the highest thermal and electrical conductivity of any known material. Fullerenes have been used in a variety of applications, including solar cells, medical treatments and lubricants.
- The structure of graphite and fullerene are very different. Graphite consists of layers of carbon atoms arranged in a hexagonal lattice, while fullerene consists of spherical molecules made up of 60 or more carbon atoms.
- Graphite and fullerene have many applications. Graphite is used in pencils, lubricants and as a conductor in electrical components, while fullerenes have been used in solar cells, medical treatments and lubricants.

Nanomaterials; Introduction, Preparation, characteristics of nanomaterials and applications

Nanomaterials are materials which have at least one dimension in the nanoscale range (1-100 nanometers). They are typically created by a variety of methods, such as chemical synthesis, physical deposition, or self-assembly.

- Introduction: Nanomaterials have unique physical and chemical properties compared to their larger counterparts. These properties make them desirable for a variety of applications, such as electronics, medical diagnostics, and drug delivery.
- Preparation: Nanomaterials can be prepared using a variety of methods, such as chemical synthesis, physical deposition, or self-assembly. Chemical synthesis involves the use of precursors to create a desired nanomaterial, while physical deposition involves the use of a physical force to deposit nanomaterials onto a substrate. Self-assembly involves the use of molecular interactions to form a desired structure.
- Characteristics of Nanomaterials: Nanomaterials have unique physical and chemical characteristics due to their small size. These characteristics include high surface area-to-volume ratio, high reactivity, and quantum confinement. These properties make them desirable for a variety of applications.
- Applications: Nanomaterials have a wide range of potential applications, including electronics, medical diagnostics, drug delivery, and energy storage. They are also being explored for use in environmental remediation, water purification, and catalysis.

Carbon Nano Tubes (CNTs)

CNTs are nanomaterials composed of rolled sheets of carbon atoms. They are hollow cylinders with diameters of 1-50 nanometers and lengths of up to several millimeters. CNTs have unique properties that make them attractive for use in a variety of applications, including electronics, energy storage, and biomedical devices.

1. Structure: CNTs are composed of a single layer of carbon atoms arranged in a hexagonal lattice. This structure gives CNTs a high degree of strength and flexibility.
2. Properties: CNTs have a high electrical conductivity, thermal conductivity, and surface-to-volume ratio, making them ideal for use in electronics and energy storage applications. They also have a high degree of chemical stability and are resistant to corrosion.
3. Applications: CNTs have been used in a variety of applications, including electronics, energy storage, and biomedical devices. They have been used as field emitters in flat panel displays, as electrodes in lithium-ion batteries, and as sensors in medical devices.

Green Chemistry: Introduction, 12 Principles and Importance of Green Synthesis

Green chemistry is a branch of chemistry that focuses on the development of processes and products that reduce or eliminate the use and generation of hazardous substances. It is a set of principles that guides chemists to design and develop chemical processes and products that minimize or eliminate the use or generation of hazardous substances.

The 12 principles of green chemistry are:

1. Prevention: It is better to prevent waste than to treat or clean up waste after it has been created.
2. Atom Economy: Synthetic methods should be designed to maximize the incorporation of all materials used in the process into the final product.
3. Less Hazardous Chemical Syntheses: Whenever practicable, synthetic methods should be designed to use and generate substances that possess little or no toxicity to human health and the environment.
4. Designing Safer Chemicals: Chemical products should be designed to effect their desired function while minimizing their toxicity.
5. Safer Solvents and Auxiliaries: The use of auxiliary substances (e.g. solvents, separation agents, etc.) should be made unnecessary wherever possible and innocuous when used.
6. Design for Energy Efficiency: Energy requirements of chemical processes should be recognized for their environmental and economic impacts and should be minimized.

7. Use of Renewable Feedstocks: A raw material or feedstock should be renewable rather than depleting whenever technically and economically practicable.
8. Reduce Derivatives: Unnecessary derivatization (use of blocking groups, protection/ deprotection, temporary modification of physical/chemical processes) should be minimized or avoided if possible, as such steps require additional reagents and can generate waste.
9. Catalysis: Catalytic reagents (as selective as possible) are superior to stoichiometric reagents.
10. Design for Degradation: Chemical products should be designed so that at the end of their function they break down into innocuous degradation products and do not persist in the environment.
11. Real-time Analysis for Pollution Prevention: Analytical methodologies need to be further developed to allow for real-time, in-process monitoring and control prior to the formation of hazardous substances.
12. Inherently Safer Chemistry for Accident Prevention: Substances and the form of a substance used in a chemical process should be chosen to minimize the potential for chemical accidents, including releases, explosions, and fires.

The importance of green synthesis lies in the fact that it reduces the use of hazardous materials and reduces the amount of waste generated. It also helps to reduce the cost of production and improves the efficiency of the process. Green synthesis also helps to reduce the environmental impact of the process and makes it more sustainable.

Chemicals, Synthesis of typical organic compounds by conventional and Green route (Adipic)

- Organic compounds can be synthesized by conventional methods, which typically involve the use of toxic reagents and solvents, and often produce large amounts of waste.
- Green synthesis is an alternative method of synthesizing organic compounds that uses fewer hazardous reagents and solvents, and produces less waste.
- Adipic acid is an example of an organic compound that can be synthesized by both conventional and green methods.
- Conventional synthesis of adipic acid involves the oxidation of cyclohexanol with nitric acid, followed by hydrolysis of the resulting cyclohexanone oxime.
- Green synthesis of adipic acid involves the oxidation of cyclohexanol with hydrogen peroxide, followed by hydrolysis of the resulting cyclohexanone oxime.
- Both conventional and green syntheses of adipic acid produce the same product, but the green route is more eco-friendly and produces less waste.

Acid and Paracetamol

- Acid is a type of chemical compound that has a sour taste and reacts with bases to form salts. Acids are found in many everyday items, such as vinegar, citrus fruits, and antacids.
- Paracetamol is an analgesic drug commonly used to relieve pain and reduce fever. It is also known as acetaminophen and is available over-the-counter in most countries.

Environmental Impact of Green Chemistry

- Green chemistry is a set of principles and practices that aim to reduce the environmental impact of the production and use of chemicals.
- Green chemistry focuses on reducing the amount of hazardous substances produced, reducing energy consumption, and reducing waste.
- Green chemistry also promotes the use of renewable resources instead of petrochemicals and encourages the use of more benign solvents and reaction conditions.
- The environmental impact of green chemistry is significant, as it can reduce the amount of hazardous waste produced, reduce energy consumption, and reduce the amount of pollutants released into the environment.

Unit-2: 8

1. Unit-2: 8 is a term used to describe a set of principles and practices that are intended to create an inclusive and respectful environment for all people.
2. This includes recognizing and respecting the diversity of individuals, their backgrounds, and their experiences.
3. Unit-2: 8 also emphasizes the importance of creating an environment that is free from discrimination, harassment, and other forms of prejudice.
4. The goal of Unit-2: 8 is to create an environment in which all people feel safe, respected, and valued.
5. To achieve this, Unit-2: 8 encourages people to be aware of their own biases and to be mindful of how their words and actions can affect others.
6. Unit-2: 8 also encourages people to be open to learning about different cultures, perspectives, and experiences.
7. By creating a culture of inclusivity, respect, and understanding, Unit-2: 8 helps to foster an environment in which all people can thrive.

Spectroscopic Techniques and Applications: Elementary idea and Simple Applications

- Spectroscopy is the study of the interaction between matter and electromagnetic radiation. It is used to determine the structure and composition of materials.

- The most common spectroscopic techniques are absorption, emission, and reflection spectroscopy.
- Absorption spectroscopy is used to study the absorption of light by a material and is used to determine the concentration of certain elements in a sample.
- Emission spectroscopy is used to study the emission of light from a material and is used to determine the composition of a sample.
- Reflection spectroscopy is used to study the reflection of light from a material and is used to determine the surface properties of a sample.
- Spectroscopy can also be used to study the structure of molecules, the motion of particles, and the properties of matter at a microscopic level.
- Spectroscopic techniques have numerous applications in fields such as medicine, chemistry, biology, and engineering. For example, spectroscopy can be used to detect and identify chemicals in the environment, to study the structure of proteins and other biological molecules, and to study the properties of materials.

UV, IR and NMR, Numerical Problems

- Ultraviolet (UV) spectroscopy is a technique used to study the structure and composition of molecules. It involves the use of light in the ultraviolet region of the electromagnetic spectrum, which is absorbed by molecules in the sample, causing them to vibrate. The vibrations are then detected and used to identify the molecules present in the sample.
- Infrared (IR) spectroscopy is a technique used to study the structure and composition of molecules. It involves the use of light in the infrared region of the electromagnetic spectrum, which is absorbed by molecules in the sample, causing them to vibrate. The vibrations are then detected and used to identify the molecules present in the sample.
- Nuclear Magnetic Resonance (NMR) spectroscopy is a technique used to study the structure and composition of molecules. It involves the use of a magnetic field to align the nuclei of atoms in a sample, and then detecting the energy released when the nuclei return to their original alignment. The energy released is used to identify the molecules present in the sample.
- Numerical problems are mathematical problems that require numerical solutions. These problems can involve equations, inequalities, systems of equations, and other mathematical concepts. Solutions to these problems involve finding the numerical values of unknown variables, or the numerical values of certain parameters.

Stereochemistry: Optical isomerism in compounds without chiral carbon, Geometrical

- Geometrical isomerism is a type of optical isomerism that occurs in compounds without chiral carbon atoms.
- This type of isomerism is caused by restricted rotation around a single bond.
- Geometrical isomers differ in their spatial arrangement due to the presence of a double bond or ring structure that prevents rotation of the atoms.
- Geometrical isomers are identified by the prefixes *cis* and *trans*, which indicate the relative positions of the two groups attached to the double bond or ring.
- The two isomers of a compound can differ in physical properties such as boiling point, melting point, solubility, and refractive index.
- Geometrical isomerism is common in compounds containing a double bond between two carbon atoms, but it can also occur in compounds containing other elements such as nitrogen, sulfur, and phosphorus.

Isomerism, Chiral Drugs

1. Isomerism is a form of chemical structure in which two or more compounds have the same molecular formula but different structural formulas.
2. Chiral drugs are molecules that have the same molecular formula but different spatial arrangements, resulting in two forms that are mirror images of each other.
3. These molecules are known as enantiomers, and they can have different effects on the body depending on which form is ingested.
4. Chiral drugs are often used in medicine to treat a variety of conditions, as they can be more effective than non-chiral drugs.
5. Enantiomers can be separated using a variety of methods, including chromatography and crystallization.
6. It is important to note that chiral drugs can have different effects on the body depending on which enantiomer is ingested, so it is important to ensure that the correct form is administered.

Unit-3: 8

1. CommonMark Markdown is a lightweight markup language that is used to format text. It is a popular choice for creating documents, webpages, and other text-based content.
2. CommonMark Markdown is designed to be easy to read and write. It uses simple formatting syntax to make text look more organized and presentable.
3. CommonMark Markdown is used to create documents that are both human-readable and machine-readable. It allows for a more structured

format for text and makes it easier to parse data.

4. CommonMark Markdown is used to create documents that are both portable and future-proof. It is designed to be backward compatible and can be used on any platform or device.
5. CommonMark Markdown is a great choice for creating documents that are both professional-looking and easy to understand. It is a popular choice for creating documents for both personal and professional use.

Electrochemistry and Batteries: Basic Concepts of Electrochemistry

1. Electrochemistry is the branch of chemistry that studies the relationship between electrical energy and chemical reactions.
2. A battery is a device that stores energy in the form of chemical energy and releases it as electrical energy.
3. A galvanic cell is the basic unit of a battery. It consists of two half-cells, each containing an electrode and an electrolyte.
4. The anode is the electrode where oxidation occurs and electrons are released. The cathode is the electrode where reduction occurs and electrons are absorbed.
5. The electrolyte is a solution that contains ions and allows them to move between the electrodes.
6. The electric current is produced by the movement of ions from the anode to the cathode.
7. The voltage of a battery is determined by the difference in the standard reduction potentials of the two half-cells.
8. The capacity of a battery is determined by the amount of energy it can store and release.
9. Rechargeable batteries can be recharged by applying an external voltage.
10. Batteries are used in a wide variety of applications, such as powering electronic devices and electric vehicles.

Batteries; Classification and applications of Primary Cells (Dry Cell) and Secondary Cells

- Batteries are devices that store and convert chemical energy into electrical energy.
- Primary cells, also known as dry cells, are batteries that cannot be recharged and are used once. Examples of primary cells include alkaline batteries, carbon-zinc batteries, and lithium batteries.
- Secondary cells are batteries that can be recharged and used multiple times. Examples of secondary cells include lead-acid batteries, nickel-cadmium batteries, and lithium-ion batteries.
- Primary cells are used in many everyday devices, such as flashlights, remote controls, and clocks.

- Secondary cells are used in devices that require a large amount of energy, such as electric vehicles, power tools, and UPS systems.

Lead Acid Battery

Lead acid batteries are one of the oldest and most commonly used types of rechargeable batteries. They are used in a wide variety of applications, including automobiles, marine vessels, backup power supplies, and even some consumer electronics.

Lead acid batteries are composed of lead plates, an electrolyte solution, and a container. The lead plates are the positive and negative electrodes, and the electrolyte solution serves as the medium through which current is conducted.

Lead acid batteries have a number of advantages. They are relatively inexpensive compared to other types of batteries, and they are capable of withstanding deep discharge cycles without significant damage. They also have a relatively long service life if properly maintained.

However, lead acid batteries also have some drawbacks. They are relatively heavy and bulky, and they require periodic maintenance in order to remain in good condition. They also have a relatively low energy density, meaning that they are not as efficient as some other types of batteries.

In conclusion, lead acid batteries are a reliable and cost-effective option for many applications. They are capable of withstanding deep discharge cycles and require minimal maintenance, but they are also relatively heavy and bulky and have a low energy density.

Corrosion: Introduction to Corrosion

Corrosion is the destruction of materials by chemical or electrochemical reaction with their environment. It is a natural process that occurs in all materials exposed to the environment, including metals, concrete, and other materials. Corrosion can cause a variety of problems, ranging from aesthetic damage to structural failure.

Types of Corrosion

Corrosion can be divided into two main categories: general corrosion and localized corrosion. General corrosion is a uniform attack on the entire surface of a material, while localized corrosion is a more specific attack on a certain area of the material.

- General corrosion: General corrosion is the most common type of corrosion and is a uniform attack on the surface of a material. It is caused by the presence of oxygen, water, and other chemicals in the environment.

- **Localized corrosion:** Localized corrosion is a more specific attack on a certain area of the material. It is caused by different factors than general corrosion, such as chloride ions, high temperatures, and high concentrations of certain chemicals.

Causes of Corrosion

Corrosion is caused by a variety of factors, including the presence of oxygen, water, and other chemicals in the environment. Other factors that can cause corrosion include high temperatures, high concentrations of certain chemicals, and the presence of chloride ions.

Prevention of Corrosion

The most effective way to prevent corrosion is to use materials that are resistant to corrosion, such as stainless steel, aluminum, and other alloys. Other methods of corrosion prevention include coating the material with a protective layer, using cathodic protection, and using inhibitors.

Prevention and Control of Corrosion

- Corrosion is the gradual destruction of materials (usually metals) by chemical and/or electrochemical reaction with their environment.
- Corrosion can be prevented or controlled by using corrosion-resistant materials, such as stainless steel, or by applying protective coatings, such as paint.
- Corrosion can also be prevented or controlled by using inhibitors, such as inhibitors that form a protective film on the surface of the metal.
- Corrosion can also be prevented or controlled by using cathodic protection, which involves using an external electrical current to reduce the corrosion rate of the metal.
- Corrosion can also be prevented or controlled by using sacrificial anodes, which are materials that are more easily corroded than the metal that is being protected.
- Corrosion can also be prevented or controlled by using coatings, such as galvanizing, which involves coating the metal with a layer of zinc.
- Corrosion can also be prevented or controlled by using environmental control, such as controlling the temperature and humidity of the environment.

Issues in Power Generation and Chemical Industries

1. **Power Generation:**
 - The increasing demand for electricity has led to a need for more efficient and reliable power generation methods.

- There is an increased risk of power outages due to extreme weather conditions, which can have a significant impact on businesses and households.
 - The cost of generating electricity is increasing due to the need for more efficient power plants and the rising cost of fuel.
 - The environmental impact of power generation is a major concern, as the burning of fossil fuels releases harmful pollutants into the atmosphere.
2. Chemical Industries:
 - Chemical industries face a number of environmental challenges, including air and water pollution, hazardous waste disposal, and the release of toxic chemicals into the environment.
 - The production of chemicals can be energy-intensive and can result in high levels of greenhouse gas emissions.
 - The use of chemicals in manufacturing processes can lead to the contamination of water sources, land, and air.
 - The improper disposal of chemicals can lead to long-term environmental damage.

Processing Industry, Oil & Gas Industry and Pulp & Paper Industries

1. Processing Industry: This industry encompasses a broad range of activities that involve the transformation of raw materials into products that are ready for consumption. This includes activities such as refining, packaging, and distribution.
2. Oil & Gas Industry: This industry involves the extraction, refining, and production of oil and gas. It includes activities such as exploration, drilling, and production of oil and gas.
3. Pulp & Paper Industry: This industry involves the production of paper and other paper-based products. It includes activities such as the production of pulp and paper, bleaching, and the production of paper products.

Chemistry of Engineering Materials:

- Engineering materials are materials that have been developed and engineered to meet specific requirements or objectives.
- These materials are typically designed to be strong, lightweight, and/or cost-effective.
- The chemical composition of engineering materials plays a major role in determining their properties and performance.
- Common types of engineering materials include metals, ceramics, polymers, and composites.

- Metals are materials composed of elements such as iron, aluminum, and copper.
- Ceramics are materials composed of inorganic compounds such as silicates, oxides, and carbides.
- Polymers are materials composed of long chains of molecules and include plastics and rubbers.
- Composites are materials composed of two or more distinct materials that are combined to create a material with improved properties.
- The chemical composition of engineering materials affects their strength, hardness, ductility, and other physical properties.
- The chemical composition also affects the materials' thermal, electrical, and magnetic properties.
- The chemical composition of engineering materials also affects their corrosion resistance and other chemical properties.

Cement; Constituents, Manufacturing, Hardening and Setting, Deterioration of Cement, Plaster

- Cement is a construction material made of a mixture of materials such as limestone, clay, iron oxide, and silica, which are heated and ground together.
- Cement is used to bind other materials together and is the main ingredient of concrete.
- The main constituents of cement are calcium oxide, silicon oxide, aluminum oxide, and iron oxide.
- Cement is manufactured by heating the raw materials to high temperatures in a kiln and then grinding them into a fine powder.
- Cement hardens when water is added to it and sets when it has hardened.
- Deterioration of cement can be caused by the presence of sulfates and chlorides in the environment, as well as by the presence of water.
- Plaster is a material made of a mixture of cement, sand, and water, which is used to coat walls and ceilings.

Paris (POP)

- Paris (POP) is an international agreement signed by 196 countries in December 2015 to combat climate change and its effects.
- It is the first legally binding universal agreement on climate change, which aims to reduce global emissions and keep global temperature rise this century well below 2°C.
- The agreement sets out a global action plan to reduce emissions and adapt to the impacts of climate change. It also includes provisions for developing countries to receive financial and technological support to adapt to climate change.
- The agreement also includes a global stocktake every five years to assess

collective progress towards achieving the long-term goal of the agreement.

- The Paris Agreement is an important step forward in the global effort to tackle climate change and its impacts.

Unit-4: 8

1. Markdown is a lightweight markup language used to format plain text. It is designed to be easy to read and write, and is often used for writing documentation for software projects.
2. Markdown is commonly used in web publishing, and is often used to create rich text documents for online publishing. It is also used to format text for email and other messaging applications.
3. Markdown is written in plain text, and can be converted to HTML or other markup languages. It is often used as a way to create formatted text quickly, without having to write HTML or other markup languages.
4. Markdown is designed to be easy to read and write, and is often used for writing documentation for software projects. It is also used to format text for email and other messaging applications.
5. Markdown is often used to create rich text documents for online publishing. It is also used to format text for email and other messaging applications.
6. Markdown is written in plain text, and can be converted to HTML or other markup languages. It is often used as a way to create formatted text quickly, without having to write HTML or other markup languages.
7. Markdown is designed to be extensible, and allows users to create custom syntax elements. This makes it possible to create custom formatting for specific types of content, such as code blocks, tables, and lists.
8. Markdown is an open source project, and is widely supported by a variety of software applications and websites. It is also supported by many text editors, making it easy to write and edit Markdown documents.

Water Technology: Sources and impurities of water, Hardness of water, Boiler troubles

- Sources of water: Water can be obtained from a variety of sources, including rivers, lakes, groundwater, and aquifers. The quality of water from each source varies and can be affected by environmental factors, such as pollution and natural events.
- Impurities of water: Water can contain a variety of impurities, including minerals, bacteria, and chemicals. These impurities can affect the taste, odor, and color of the water, as well as its suitability for use in various applications.
- Hardness of water: Hardness of water is a measure of the amount of calcium and magnesium present in the water. Hard water can cause scaling in pipes and fixtures, and can affect the effectiveness of soaps and detergents.

- **Boiler troubles:** Boilers are used to heat water for a variety of applications. If the water is too hard, it can cause scaling and other problems in the boiler, which can lead to reduced efficiency and increased maintenance costs.

Techniques for Water Softening

- **Lime-Soda:** This method is also known as the soda-ash process and involves adding lime (calcium hydroxide) and soda (sodium carbonate) to water. This process increases the pH of the water, which causes the hardness-causing ions to precipitate out.
- **Zeolite:** Zeolite is a naturally occurring mineral that is used to soften water. It works by exchanging sodium ions for calcium and magnesium ions, thus removing the hardness-causing ions from the water.
- **Ion Exchange:** This process involves passing water through a resin bed that contains sodium ions. The hardness-causing ions are exchanged for sodium ions, thus softening the water.
- **Reverse Osmosis:** This process involves passing water through a semi-permeable membrane. The membrane traps the hardness-causing ions, thus softening the water.

Process), Determination of Hardness and Alkalinity, Numerical Problems

1. **Hardness:** Hardness is a measure of the amount of calcium and magnesium ions present in water. It is expressed in ppm (parts per million) or mg/L (milligrams per liter).
2. **Alkalinity:** Alkalinity is a measure of the buffering capacity of water. It is expressed in ppm (parts per million) or mg/L (milligrams per liter).
3. **Numerical Problems:** Numerical problems involve solving equations or using numerical methods to solve problems. These problems can be solved using calculators or computers. Examples of numerical problems include finding the roots of a polynomial equation, solving a system of linear equations, and finding the minimum or maximum of a function.

Fuels and Combustion: Definition, Classification, Characteristics of a good fuel, Calorific

- **Definition:** A fuel is any material that can be used to produce energy, usually in the form of heat. Combustion is the process of burning a fuel in the presence of oxygen to produce energy.

- **Classification:** Fuels are generally classified according to their composition (e.g. solid, liquid, or gaseous) and their source (e.g. fossil fuels or renewable sources).
- **Characteristics of a good fuel:** A good fuel should be easy to ignite, produce a large amount of energy, and have a low environmental impact.
- **Calorific:** The calorific value of a fuel is a measure of the amount of energy it can produce when burned.

Values, Gross & Net Calorific Value, Determination of Calorific Value by Bomb Calorimeter

- Calorific value is the amount of energy released when a material is burned.
- The gross calorific value is the total energy released when a material is burned in oxygen, while the net calorific value is the energy released when the material is burned in air.
- The determination of calorific value is done using a bomb calorimeter. This device has a pressure-tight container, called a bomb, which is filled with oxygen.
- The material to be tested is placed in the bomb and then ignited. The heat released by the burning material is absorbed by a known mass of water in the calorimeter, and the change in temperature is measured.
- The calorific value of the material is then calculated using the equation:

Calorific Value = (Mass of Water x Specific Heat Capacity x Change in Temperature) / Mass of Material

- Calorific values are used to compare the energy content of different materials and to determine the efficiency of combustion processes.

Theoretical Calculation of Calorific Value by Dulong's Method

Calorific value is the amount of energy released when a unit mass of fuel is burned. Dulong's method is a simple theoretical method used to calculate the calorific value of a fuel from its elemental composition. This method is based on the assumption that the calorific value of a fuel is the sum of the calorific values of its constituent elements. The formula for calculating the calorific value using Dulong's method is as follows:

Calorific value (in kcal/kg) = 3.7 x Carbon content + 8.8 x Hydrogen content + 25.2 x Oxygen content + 11.3 x Sulphur content

Ranking of Coal

Coal is ranked based on its calorific value, volatile matter content, ash content and moisture content. The higher the calorific value, the higher the ranking of the coal. The calorific value is determined by Dulong's method, as described above.

Analysis of Coal

The analysis of coal involves the determination of its calorific value, volatile matter content, ash content and moisture content. The calorific value is determined by Dulong's method, as described above. The volatile matter content is determined by heating the coal sample in an oven at a temperature of 950°C . The ash content is determined by heating the coal sample in an oven at a temperature of 950°C and then measuring the mass of the ash residue. The moisture content is determined by heating the coal sample in an oven at a temperature of 105°C and then measuring the mass of the water evaporated.

Proximate and Ultimate Analysis Method

Proximate analysis is a method of determining the approximate percentage of the main components of a substance, such as ash, moisture, volatile matter, and fixed carbon. Ultimate analysis is a method of determining the exact percentage of the main components of a substance, such as carbon, hydrogen, nitrogen, and sulfur.

Numerical Problems

Numerical problems are mathematical problems that involve the use of numerical data. These problems can involve basic arithmetic operations, such as addition, subtraction, multiplication, and division, or more advanced operations, such as calculus, trigonometry, and linear algebra.

Chemistry of Biogas

Biogas is a type of renewable energy produced from the decomposition of organic matter. The main components of biogas are methane (CH_4) and carbon dioxide (CO_2). The chemistry of biogas involves the reaction of these components with other elements in the environment, such as oxygen, to produce energy. The reaction of methane and oxygen is known as the methanogenesis reaction, which is the main source of energy produced by biogas.

Production from Organic Waste Materials and Their Environmental Impact on Society

1. Organic waste materials are materials that are produced from plants and animals, such as food scraps, yard trimmings, and animal manure.
2. Organic waste materials can be processed into a variety of products, such as compost, fuel, and bioplastics.
3. Composting organic waste materials helps reduce the amount of waste sent to landfills, and can improve soil fertility.
4. Burning organic waste materials can produce energy, but can also release air pollutants into the atmosphere.
5. Producing bioplastics from organic waste materials can help reduce the amount of plastic waste that is sent to landfills.
6. Using organic waste materials to produce energy and other products can help reduce the amount of greenhouse gases emitted into the atmosphere.
7. Organic waste materials can also be used to produce biogas, which can be used for cooking and other energy needs.
8. Improper disposal of organic waste materials can lead to pollution of water, air, and soil, as well as the spread of disease.

Unit-5: 8

1. Unit-5: 8 is a unit of study focusing on the principles of information security.
2. It covers topics such as risk management, access control, authentication, encryption, and data integrity.
3. Risk management is the process of assessing the potential risks associated with an information system and taking steps to reduce those risks.
4. Access control is the process of restricting access to information systems and data, ensuring that only authorized users can access them.
5. Authentication is the process of verifying the identity of a user or system before granting access to a system or data.
6. Encryption is the process of transforming data into a form that cannot be read by unauthorized users, ensuring data confidentiality.
7. Data integrity is the process of ensuring that data has not been tampered with or corrupted during transmission or storage.
8. Unit-5: 8 is an important unit of study for anyone interested in information security.

Materials Chemistry:

- Materials Chemistry is the study of the structure and properties of materials and their interactions with the environment.
- It is an interdisciplinary field that draws on concepts from physics, chemistry, engineering, and biology to develop new materials with desired properties.
- Materials Chemistry is concerned with the synthesis, processing, and characterization of materials, as well as the study of their structure and properties.
- Understanding the structure-property relationships of materials is essential for the development of new materials with desired properties.
- It is also important for the understanding of existing materials, and for the optimization of their properties for specific applications.
- In Materials Chemistry, the structure of materials is studied at multiple length scales, from the atomic to the macroscopic level.
- This includes understanding the relationships between the atomic structure, the microstructure, and the macroscopic properties of materials.
- The properties of materials can be altered by changing their composition, structure, or processing.
- Materials Chemistry is also concerned with the development of new materials, such as nanomaterials, and with the study of the behavior of materials in extreme environments, such as high temperatures and pressures.

Polymers; Classification, Polymerization processes, Thermosetting and Thermoplastic

Polymers are large molecules composed of repeating structural units. These molecules can be natural or synthetic and can vary in size and complexity. Polymers are classified according to the type of monomer units they are composed of and the type of polymerization process used to create them.

- Classification:
 - Natural polymers: These polymers are found in nature and are composed of monomers that are linked together by covalent bonds. Examples include proteins, DNA, cellulose, and rubber.
 - Synthetic polymers: These polymers are created in the laboratory from monomers that are linked together by covalent bonds. Examples include polyethylene, polystyrene, and nylon.
- Polymerization Processes:
 - Addition polymerization: This is a process in which monomers are joined together by covalent bonds to form a polymer. This process can be catalyzed by either an acid or a base.
 - Condensation polymerization: This is a process in which two monomers are joined together by covalent bonds to form a polymer. This process is usually catalyzed by an acid or a base.
- Thermosetting and Thermoplastic Polymers:

- Thermosetting polymers: These polymers are formed by the condensation polymerization of monomers. Once these polymers have been formed, they cannot be reshaped or reformed. Examples include epoxy and polyurethane.
- Thermoplastic polymers: These polymers are formed by the addition polymerization of monomers. These polymers can be reshaped and reformed by applying heat. Examples include polyethylene and polystyrene.

Polymers

Polymers are large molecules made up of repeating structural units connected by covalent chemical bonds. They can be either synthetic or naturally occurring and are characterized by their ability to be molded into various shapes and sizes. Synthetic polymers are used in a wide variety of applications, including adhesives, plastics, coatings, fibers, and elastomers.

Polymer Blends and Composites

Polymer blends and composites are materials composed of two or more polymers, or a polymer and a non-polymeric material. These materials are used in a variety of applications, including automotive and aerospace components, medical devices, and consumer products. The properties of the material can be tailored to specific needs by varying the composition of the blend or composite.

Conducting Polymers

Conducting polymers are polymers that can conduct electricity. These materials are used in a variety of applications, including organic electronics, sensors, and energy storage devices. They are typically made from conjugated polymers chains, which contain alternating single and double bonds along the backbone of the polymer.

Biodegradable Polymers

Biodegradable polymers are polymers that can be broken down by microorganisms into simpler molecules. These materials are used in a variety of applications, including packaging and medical devices. They are typically made from polymers that contain hydrolyzable linkages, which can be broken down into smaller molecules by the action of water.

Preparation, Properties, and Industrial Applications of Teflon, Lucite, Bakelite, Kelvar, and Dacron

- **Teflon:** Teflon is a synthetic fluoropolymer of tetrafluoroethylene. It is a strong, non-reactive material that is used in a variety of applications due to its chemical resistance and low coefficient of friction. Teflon is prepared by polymerizing tetrafluoroethylene in aqueous suspension. Its properties include high melting point, low coefficient of friction, and high thermal stability. Teflon is used in a variety of industrial applications, including as a coating for cookware, as a lubricant for machinery, and as a non-stick coating for fabrics.
- **Lucite:** Lucite is a brand name for a type of acrylic resin. It is a transparent, lightweight material with excellent optical properties. Lucite is prepared by polymerizing methyl methacrylate. Its properties include high optical clarity, low weight, and excellent weather-resistance. Lucite is used in a variety of industrial applications, including as a coating for aircraft and automotive parts, as a protective coating for furniture, and as a material for signage.
- **Bakelite:** Bakelite is a type of thermosetting plastic. It is a strong, hard material that is resistant to heat, fire, and chemicals. Bakelite is prepared by polymerizing phenol formaldehyde. Its properties include high strength and rigidity, low electrical conductivity, and excellent chemical resistance. Bakelite is used in a variety of industrial applications, including as a material for electrical insulation, as a molding compound for castings, and as a protective coating for machinery.
- **Kelvar:** Kelvar is a brand name for a type of aramid fiber. It is a strong, lightweight material with excellent thermal and chemical resistance. Kelvar is prepared by spinning a solution of poly-paraphenylene terephthalamide. Its properties include high tensile strength, low weight, and excellent heat resistance. Kelvar is used in a variety of industrial applications, including as a material for protective clothing, as a reinforcement for composites, and as a material for bullet-proof vests.
- **Dacron:** Dacron is a brand name for a type of polyester fiber. It is a strong, lightweight material with excellent thermal and chemical resistance. Dacron is prepared by spinning a solution of polyethylene terephthalate. Its properties include high tensile strength, low weight, and excellent heat resistance. Dacron is used in a variety of industrial applications, including as a material for clothing, as a reinforcement for composites, and as a material for sails and boat covers.

Thiokol

Thiokol is a synthetic rubber material that was developed in the 1930s. It is used in a variety of applications, including tires, adhesives, and lubricants. Thiokol is known for its durability and resistance to weathering. It is also resistant to oil and gasoline.

However, Thiokol has been found to be a source of air pollution. It releases volatile organic compounds (VOCs) into the atmosphere, which can contribute to global warming and smog formation.

Nylon

Nylon is a synthetic polymer material that was developed in the 1930s. It is used in a variety of applications, including clothing, carpets, and industrial materials. Nylon is known for its strength, durability, and resistance to wear and tear. It is also resistant to mildew and mold.

However, the production of nylon can be a source of air pollution. It releases nitrogen oxides (NO_x) into the atmosphere, which can contribute to global warming and smog formation.

Buna-N

Buna-N is a synthetic rubber material that was developed in the 1930s. It is used in a variety of applications, including tires, adhesives, and lubricants. Buna-N is known for its durability and resistance to weathering. It is also resistant to oil and gasoline.

However, Buna-N has been found to be a source of air pollution. It releases volatile organic compounds (VOCs) into the atmosphere, which can contribute to global warming and smog formation.

Buna-S

Buna-S is a synthetic rubber material that was developed in the 1930s. It is used in a variety of applications, including tires, adhesives, and lubricants. Buna-S is known for its durability and resistance to weathering. It is also resistant to oil and gasoline.

However, Buna-S has been found to be a source of air pollution. It releases volatile organic compounds (VOCs) into the atmosphere, which can contribute to global warming and smog formation.

Environmental Impact

The production and use of synthetic rubber materials such as Thiokol, Nylon, Buna-N and Buna-S can have a negative impact on the environment. The release of volatile organic compounds (VOCs) and nitrogen oxides (NO_x) into the atmosphere can contribute to global warming and smog formation. In addition, the production of these materials can lead to the release of toxic chemicals into the environment, which can have an adverse effect on human health and the environment.

Polymers

Polymers are large molecules composed of repeating structural units. These molecules are characterized by their high molecular mass, which can range from a few hundred to several million. Polymers can be found in a variety of products, including plastics, rubber, adhesives, fabrics, and even pharmaceuticals.

1. Types of Polymers
 - Natural Polymers: Natural polymers are those that are found in nature, such as proteins, wood, and silk.
 - Synthetic Polymers: Synthetic polymers are those that are created by humans, such as plastics and synthetic rubber.
2. Structure of Polymers
 - Monomers: Monomers are the small molecules that are linked together to form polymers.
 - Polymerization: Polymerization is the process in which monomers are linked together to form polymers.
3. Properties of Polymers
 - Flexibility: Polymers are flexible due to their long chain structure, which allows them to bend and stretch.
 - Strength: Polymers can be strong, depending on the type of monomer used and the way the polymer is formed.
 - Durability: Polymers are generally durable and can withstand high temperatures, which makes them ideal for a variety of applications.

Organometallic Compounds: General methods of Preparation and Applications

- Organometallic compounds are molecules that contain at least one metal-carbon bond.
- They are generally prepared by reacting a metal with an organic compound.
- Common methods of preparation include metathesis reactions, alkyl-metal exchange, and metal-halogen exchange.
- Common applications of organometallic compounds include catalysts for organic synthesis, reagents for organic transformations, and ligands for

coordination complexes.

- Organometallic compounds are also used in the pharmaceutical and petrochemical industries.

Organometallic compounds (RMgX and LiAlH₄)

Organometallic compounds are a class of compounds that contain a metal-carbon bond. The most common organometallic compounds are RMgX and LiAlH₄.

RMgX

RMgX compounds are a type of organometallic compound that contains a metal-carbon bond. The metal is typically a transition metal, such as ruthenium, rhodium, palladium, osmium, or iridium. The carbon is typically an alkyl group, such as methyl, ethyl, or propyl.

LiAlH₄

LiAlH₄ is an organometallic compound that contains a metal-carbon bond. The metal is lithium and the carbon is an alkyl group, typically an alkyl halide. LiAlH₄ is used in organic synthesis as a reducing agent. It can reduce aldehydes, ketones, carboxylic acids, and other organic compounds.

Course Outcomes:

- Develop an understanding of the fundamental principles of computer science and programming.
- Learn to design and implement algorithms and data structures to solve problems.
- Develop an understanding of the principles of software engineering and design.
- Develop an understanding of the principles of distributed systems and networks.
- Develop an understanding of the principles of operating systems and databases.
- Develop an understanding of the principles of artificial intelligence, machine learning, and natural language processing.
- Demonstrate the ability to develop and implement a software project.

Upon completion of the course the student should be able to:

- Identify and explain the core concepts of the subject
- Analyze and interpret data related to the subject
- Develop and apply skills related to the subject
- Apply critical thinking to solve problems related to the subject
- Analyze and evaluate the impact of the subject on society

- Demonstrate an understanding of the ethical considerations of the subject
- Communicate effectively about the subject

Units Course Outcomes Bloom's

1. Understand the fundamentals of Bloom's Taxonomy, including the six levels of cognitive learning and their associated verbs.
2. Recognize how to apply Bloom's Taxonomy to create course outcomes that are specific, measurable, achievable, relevant, and time-bound (SMART).
3. Construct course outcomes that are measurable and assessable using Bloom's Taxonomy.
4. Identify the appropriate level of cognitive learning for each course outcome.
5. Utilize Bloom's Taxonomy to create course objectives that are aligned with course outcomes.
6. Analyze the implications of Bloom's Taxonomy on the design and delivery of courses.

Level

1. Level is a measure of a person's ability or skill in a particular field or activity.
2. It is often used to refer to an individual's ability to perform a certain task, or to refer to the difficulty of a task.
3. There are different levels of ability or difficulty, such as beginner, intermediate, advanced, and expert.
4. Level can also refer to a person's rank or status within an organization or group, such as a company or school.
5. Level is also used to refer to the degree of complexity or sophistication of a system, such as a computer system.

CO-1 Get an Understanding of the Theoretical Principles of Chemistry of Molecular Structure, Bonding and

1. Molecular structure is the arrangement of atoms in a molecule. It can be determined by various methods, such as X-ray crystallography or Nuclear Magnetic Resonance (NMR).
2. Bonding is the process by which atoms are held together in a molecule. Chemical bonds can be covalent, ionic, or metallic.
3. The theory of molecular structure and bonding is based on the principles of quantum mechanics. It is used to explain the properties of molecules, such as their reactivity and spectroscopic properties.
4. The theory of molecular structure and bonding also explains the nature of intermolecular forces, such as van der Waals forces and hydrogen bonding. These forces are responsible for the physical properties of molecules, such as boiling point and melting point.

Properties, Chemistry of Advanced Materials (Liquid Crystals, Nanomaterials, Graphite & Fullerene)

1. **Liquid Crystals:** Liquid crystals are materials that exhibit an ordered arrangement of molecules in the liquid state. They are composed of organic molecules that are arranged in a specific way, allowing them to have properties of both liquids and solids. Examples of liquid crystals include the liquid crystal display (LCD) used in digital watches and televisions.
2. **Nanomaterials:** Nanomaterials are materials that are composed of particles that are less than 100 nanometers in size. These materials have unique properties due to their small size, such as increased surface area, increased reactivity, and increased strength. Examples of nanomaterials include carbon nanotubes, quantum dots, and graphene.
3. **Graphite:** Graphite is a form of carbon with a layered structure. It is a soft, dark gray solid with a metallic luster. It is used in many applications, such as lubricants, batteries, and pencils.
4. **Fullerenes:** Fullerenes are molecules composed of carbon atoms in the shape of a hollow sphere, ellipsoid, or tube. They are used in a variety of applications, such as in medicine, electronics, and nanotechnology.

Principles of Green Chemistry

1. **Prevention:** It is better to prevent waste than to treat or clean up waste after it has been created.
2. **Atom Economy:** Synthetic methods should be designed to maximize the incorporation of all materials used in the process into the final product.
3. **Less Hazardous Chemical Syntheses:** Wherever practicable, synthetic methods should be designed to use and generate substances that possess little or no toxicity to human health and the environment.
4. **Designing Safer Chemicals:** Chemical products should be designed to effect their desired function while minimizing their toxicity.
5. **Safer Solvents and Auxiliaries:** The use of auxiliary substances (e.g. solvents, separation agents, etc.) should be made unnecessary wherever possible and innocuous when used.
6. **Design for Energy Efficiency:** Energy requirements of chemical processes should be recognized for their environmental and economic impacts and should be minimized. If possible, synthetic methods should be conducted at ambient temperature and pressure.

7. Use of Renewable Feedstocks: A raw material or feedstock should be renewable rather than depleting whenever technically and economically practicable.
8. Reduce Derivatives: Unnecessary derivatization (use of blocking groups, protection/de-protection, and temporary modification of physical/chemical processes) should be minimized or avoided if possible, because such steps require additional reagents and can generate waste.
9. Catalysis: Catalytic reagents (as selective as possible) are superior to stoichiometric reagents.
10. Design for Degradation: Chemical products should be designed so that at the end of their function they break down into innocuous degradation products and do not persist in the environment.

K3

1. K3 is the name given to a type of artificial intelligence (AI) that is designed to interact with humans in a natural language. It is capable of understanding and responding to questions, commands, and requests posed in human language.
2. K3 AI systems are typically built using natural language processing (NLP) and machine learning (ML) technologies. The AI system is trained on a large corpus of human language data in order to recognize patterns and interpret human input.
3. K3 AI systems are used in a variety of applications, such as customer service, virtual assistants, and chatbots. They can help automate tasks, provide customer support, and answer questions.
4. K3 AI systems are becoming increasingly popular as they can provide more natural interactions with humans and reduce the need for manual labor. They are also able to learn and improve over time, making them more efficient and effective.

CO-2 Apply the fundamental concepts of determination of structure with various spectral techniques and

1. Carbon dioxide (CO₂) is a molecule composed of two oxygen atoms and one carbon atom.
2. The structure of CO₂ is linear, with a carbon atom in the center and two oxygen atoms at the ends.
3. Spectroscopic techniques can be used to study the structure of CO₂.
4. Infrared spectroscopy can be used to determine the bond lengths and angles of CO₂.
5. Nuclear magnetic resonance (NMR) spectroscopy can be used to determine the chemical shifts of the atoms in CO₂.

6. Mass spectrometry can be used to determine the mass of the molecules in the sample, as well as the relative abundance of different isotopes.
7. X-ray diffraction can be used to determine the structure of molecules in the solid state.
8. Ultraviolet-visible spectroscopy can be used to determine the absorption of light by molecules in solution.

Stereochemistry

Stereochemistry is a branch of chemistry that deals with the three-dimensional arrangement of atoms in molecules and their effects on chemical and physical properties. It is an important field of study as it can help to explain the behavior of molecules, including their reactivity and stability.

Stereochemistry is based on the concept of chirality, which is the ability of molecules to exist in two forms that are mirror images of each other. These forms are known as enantiomers, and they can have different chemical and physical properties.

The study of stereochemistry includes the following topics:

- Identifying chiral centers in molecules
- Understanding the different types of stereoisomers
- Examining the effects of stereoisomers on the properties of molecules
- Analyzing the optical activity of molecules
- Investigating the reaction mechanisms of stereoselective reactions
- Exploring the use of chiral catalysts in asymmetric synthesis

K4

1. K4 is a form of artificial intelligence (AI) that is capable of understanding and responding to natural language. It is also known as natural language processing (NLP). K4 is used to interact with humans and to automate tasks such as customer service, data analysis, and natural language understanding. K4 can be used to build intelligent conversational agents, virtual assistants, and chatbots.
2. K4 is based on the principles of machine learning and deep learning. It uses algorithms to analyze and interpret natural language data, allowing it to understand the context of a conversation and respond appropriately. K4 can be trained to recognize certain words, phrases, and topics and respond accordingly.
3. K4 can be used to create intelligent virtual assistants, chatbots, and conversational agents. These agents can be used to automate customer service tasks, such as answering questions and providing recommendations. They can also be used to provide data analysis and natural language understanding.

4. K4 is a powerful tool for businesses, as it can help them to improve customer service and increase efficiency. It can also help businesses to better understand their customers and the dynamics of their industry.

CO-3 Utilize the theory of construction of electrodes, batteries and fuel cells in redesigning new engineering

1. Electrodes are components of a battery or fuel cell that allow the transfer of electrons from one side of the cell to the other.
2. Batteries are devices that store electrical energy and use it to power electronic devices.
3. Fuel cells are electrochemical devices that convert chemical energy into electrical energy.
4. The construction of electrodes, batteries and fuel cells involves the selection of materials and the design of the components.
5. The selection of materials includes the choice of active materials, current collectors, separators, and electrolytes.
6. The design of the components includes the selection of the shape, size, and configuration of the components.
7. The performance of the electrodes, batteries and fuel cells is affected by the design of the components and the selection of materials.
8. The performance of the electrodes, batteries and fuel cells can be improved by redesigning the components and selecting better materials.
9. Redesigning the components and selecting better materials can lead to the development of new engineering applications.

Products and Corrosion

Corrosion is a natural process that occurs when a material is exposed to an environment that causes it to degrade. It can be caused by a variety of factors, including chemical reactions, physical contact, and environmental conditions. The following are some common causes of corrosion:

- Chemical reactions: Certain chemicals, such as acids and bases, can cause corrosion. When these chemicals come into contact with a material, they can cause it to corrode.
- Physical contact: Physical contact, such as friction or abrasion, can cause corrosion. This type of corrosion is most common in moving parts, such as bearings and gears.
- Environmental conditions: Certain environmental conditions, such as temperature and humidity, can cause corrosion. These conditions can cause materials to corrode faster or slower, depending on the material.

To control corrosion, it is important to understand the causes and develop strategies to reduce or prevent the effects of corrosion. Some common methods of

controlling corrosion include the following:

- Protective coatings: Applying a protective coating, such as paint or a sealant, can help reduce the effects of corrosion.
- Anodic protection: Anodic protection involves applying a current to a metal to reduce its tendency to corrode.
- Cathodic protection: Cathodic protection involves applying a current to a metal to reduce the rate at which it corrodes.
- Corrosion inhibitors: Corrosion inhibitors are substances that can be added to a material to reduce its tendency to corrode.
- Corrosion-resistant materials: Certain materials are more resistant to corrosion than others. By choosing corrosion-resistant materials, it is possible to reduce the effects of corrosion.

Understanding of Chemistry of Engineering Materials (Cement)

1. Cement is a material that is composed of a mixture of chemical compounds, including calcium, silicon, aluminum, and iron.
2. Cement is used in the construction of buildings and other structures because it is a strong, durable material that can withstand extreme temperatures and pressures.
3. The chemical composition of cement is determined by the type of raw materials used in its production.
4. The most common raw materials used to make cement are limestone, clay, and sand.
5. The manufacturing process of cement involves the grinding and blending of the raw materials, followed by heating the mixture in a kiln at temperatures of up to 1500°C.
6. The heated mixture is then ground down into a fine powder, which is then mixed with water to form a paste.
7. The paste is then allowed to harden and form a solid material, which is known as cement.
8. Cement is used in the construction of buildings and other structures because it is a strong, durable material that can withstand extreme temperatures and pressures.
9. Cement is also used in the production of concrete, which is a mixture of cement, water, and aggregate materials such as sand and gravel.
10. Cement and concrete are both important materials in the construction industry and are used in a variety of applications, from small-scale residential projects to large-scale commercial and industrial projects.

K3

1. K3 is a type of artificial intelligence (AI) system that is designed to be able to understand, interpret, and respond to natural language.
2. K3 systems use natural language processing (NLP) technology to understand and interpret human language, allowing them to interact with humans in a more natural way.
3. K3 systems are used in a variety of applications, including customer service, virtual assistants, chatbots, and more.
4. K3 systems are trained using large datasets of natural language data, such as conversations, articles, and other text.
5. K3 systems are able to understand the context of a conversation and respond accordingly, making them more human-like than other AI systems.
6. K3 systems are also able to learn from their interactions with humans, allowing them to become more accurate and effective over time.

CO-4 Develop understanding of the sources, impurities and hardness of water, apply the concepts of

1. Sources of Water: Water can come from a variety of sources, including rivers, lakes, streams, wells, springs, and glaciers.
2. Impurities in Water: Water may contain a variety of impurities, such as minerals, organic matter, bacteria, and other pollutants.
3. Hardness of Water: Hardness of water is determined by the amount of dissolved minerals, such as calcium and magnesium, in the water.
4. Concepts of Water Quality: Water quality is determined by a variety of factors, including pH, turbidity, conductivity, and dissolved oxygen. Water quality can be improved by treating the water with chemicals and/or filtration.

Determination of Calorific Values and Analysis of Coal

1. Calorific value is a measure of the energy content of a fuel, expressed as the amount of heat released when a given weight of fuel is completely burned.
2. The calorific value of coal is generally determined by bomb calorimetry, which is a process in which the heat released by the combustion of a sample of coal is measured in a bomb calorimeter.
3. The bomb calorimeter consists of a steel container, known as a bomb, which is surrounded by an insulated jacket. The sample of coal is placed in the bomb and ignited. As the coal burns, the heat released is measured by the calorimeter.
4. The calorific value of coal is usually expressed in units of kilojoules per kilogram (kJ/kg) or British thermal units per pound (BTU/lb).

5. In addition to determining the calorific value of coal, it is also important to analyze the coal for other properties such as moisture content, ash content, volatile matter, and fixed carbon.
6. Moisture content is a measure of the amount of water present in the coal, and is usually expressed in units of percent by weight.
7. Ash content is a measure of the amount of inorganic material present in the coal, and is usually expressed in units of percent by weight.
8. Volatile matter is a measure of the amount of hydrocarbons present in the coal that are released when the coal is heated, and is usually expressed in units of percent by weight.
9. Fixed carbon is a measure of the amount of carbon in the coal that remains after the volatile matter has been burned off, and is usually expressed in units of percent by weight.

K3

1. K3 is a type of computer programming language that is used to create applications and websites. It is a modern language that is based on object-oriented programming (OOP) principles. K3 is an open-source language, meaning that it is free to use and modify.
2. K3 is a general-purpose language and can be used to develop applications for a variety of platforms, including web, mobile, desktop, and embedded systems. It is designed to be easy to learn and use, and supports a variety of programming paradigms, including procedural, object-oriented, and functional.
3. K3 is a strongly-typed language, meaning that it requires the programmer to declare the type of data that will be used in a program. This helps to ensure that the program will run correctly and that the data will be handled properly.
4. K3 is a compiled language, meaning that it must be translated into machine language before it can be executed. This ensures that the code is efficient and that it runs quickly.
5. K3 has a large library of built-in functions and classes that can be used to simplify the development process. It also has a large community of users who can provide support and advice.

CO-5 Develop the understanding of Chemical structure of polymers and its effect on their various properties

1. Polymers are large molecules made up of repeating structural units. Each structural unit is known as a monomer.
2. Polymers can be natural or synthetic. Natural polymers are derived from plants and animals, while synthetic polymers are man-made.

3. The chemical structure of a polymer affects its physical and chemical properties.
4. The physical properties of a polymer are mainly determined by its molecular weight, its degree of crystallinity, and its glass transition temperature.
5. The chemical properties of a polymer are mainly determined by its chemical structure, its degree of crystallinity, and its glass transition temperature.
6. The degree of crystallinity of a polymer affects its mechanical properties, such as its tensile strength and modulus.
7. The glass transition temperature of a polymer affects its thermal properties, such as its heat deflection temperature and thermal conductivity.
8. The chemical structure of a polymer affects its chemical properties, such as its solubility and reactivity.

When Used as Engineering Materials: Understanding the Applications of Specific Polymers and Chemistry

1. Polymers are materials composed of large molecules made up of repeating structural units.
2. These units are connected by covalent bonds, forming a chain-like structure.
3. Polymers are used in a wide range of applications, from packaging to construction materials.
4. The properties of a polymer depend on its chemical composition and structure.
5. Polymers can be classified according to their structure, such as linear, branched, or cross-linked.
6. The physical properties of polymers, such as their strength, flexibility, and heat resistance, are affected by their chemical structure.
7. Polymers can be used to produce materials with specific properties, such as high strength, low weight, and low cost.
8. Polymers are also used in the production of coatings, adhesives, and sealants.
9. Polymers are used in the production of medical devices, such as catheters and implants.
10. Polymers are used in the production of electronics, such as semiconductors and solar cells.

Applicable Industrial Process

1. Automation: Automation is the use of machines, control systems and information technologies to reduce the need for human work in the production

of goods and services. Automation has been used in a wide variety of industries, from manufacturing to healthcare, to increase productivity and reduce costs.

2. **Robotics:** Robotics is the branch of technology that deals with the design, construction, operation, and application of robots, as well as computer systems for their control, sensory feedback, and information processing. Robotics is used in a wide variety of industries, from manufacturing to healthcare, to automate tasks that are too difficult or dangerous for humans to perform.
3. **Data Analysis:** Data analysis is the process of examining, transforming, and modeling data with the goal of discovering useful information, informing conclusions, and supporting decision-making. Data analysis is used in a wide variety of industries, from finance to healthcare, to uncover patterns and trends in data that can be used to make better decisions.
4. **Artificial Intelligence:** Artificial intelligence (AI) is the ability of a computer system to perform tasks that normally require human intelligence, such as visual perception, speech recognition, decision-making, and translation between languages. AI is used in a wide variety of industries, from finance to healthcare, to automate tasks that would otherwise require human intelligence.

K3

1. K3 is a mathematical concept that refers to the three-dimensional Euclidean space. It is the most common type of space studied in geometry and is the space of Euclidean geometry, where the distance between two points is the length of the straight line connecting them.
2. K3 is a three-dimensional space that is defined by three axes: the x-axis, the y-axis, and the z-axis. These three axes are perpendicular to each other and form a Cartesian coordinate system.
3. In K3, a point is represented by three coordinates: (x, y, z). The distance between two points is calculated by the Pythagorean theorem: $d = \sqrt{x^2 + y^2 + z^2}$.
4. K3 is used in many branches of mathematics, including algebra, calculus, and topology. It is also used in physics and engineering to describe the properties of three-dimensional objects.

Reference Books:

- **The Norton Anthology of English Literature:** This anthology is a comprehensive collection of English literature from the medieval period to the present day. It includes works from major authors such as William Shakespeare, John Milton, and Jane Austen.

- **The Oxford Companion to English Literature:** This comprehensive reference book provides a comprehensive guide to English literature, from the Anglo-Saxon period to the present day. It includes biographies of authors, critical analysis of works, and an overview of the development of English literature.
- **The Cambridge History of English Literature:** This book provides an in-depth look at the history of English literature from the medieval period to the present day. It includes a discussion of major authors, genres, and themes, as well as an analysis of the social and cultural context in which English literature was produced.
- **The Norton Anthology of American Literature:** This anthology is a comprehensive collection of American literature from colonial times to the present day. It includes works from major authors such as Walt Whitman, Emily Dickinson, and Toni Morrison.
- **The Oxford Companion to American Literature:** This comprehensive reference book provides a comprehensive guide to American literature, from the colonial period to the present day. It includes biographies of authors, critical analysis of works, and an overview of the development of American literature.

1. Engineering Chemistry by Rath & Singh, 2nd Edition, Cengage Learning India Pvt Ltd Delhi

This book provides a comprehensive overview of the fundamentals of engineering chemistry, including topics such as atomic structure, chemical bonding, chemical kinetics, thermodynamics, electrochemistry, and more. It covers the fundamentals of engineering chemistry, and provides a comprehensive introduction to the subject. The book also includes a range of examples and problems to illustrate the concepts discussed. The book is suitable for undergraduate students of engineering, as well as those studying for postgraduate degrees.

2. Engineering Chemistry by SS Dara, S Chand & Co Ltd.

1. Engineering Chemistry is a branch of chemistry that focuses on the properties and uses of chemicals in engineering applications.
2. It is an important part of the engineering curriculum, as it provides an understanding of the physical and chemical principles that govern the behavior of materials in engineering applications.
3. The topics covered in Engineering Chemistry include thermodynamics, electrochemistry, corrosion, materials science, and catalysis.
4. The book by SS Dara, S Chand & Co Ltd. covers all the topics in detail and provides a comprehensive overview of the subject.
5. It also includes a variety of examples and exercises to help students understand the concepts better.
6. The book is suitable for undergraduate engineering students and those preparing for competitive examinations.

3. Engineering Chemistry by Jain & Jain, S.Chand & Comp, New Delhi

1. Engineering chemistry is a branch of chemistry that focuses on the application of chemical principles to engineering.
2. It involves the use of chemical processes and materials to create products and services for engineering purposes.
3. It is used in a variety of industries, from the automotive industry to the aerospace industry.
4. It is also used in the medical field, where it is used to create diagnostic and therapeutic agents.
5. It is also used in the food industry, where it is used to improve the quality of food products.
6. It is also used in the environmental field, where it is used to create solutions to environmental problems.
7. It is also used in the energy industry, where it is used to create energy efficient products.
8. It is also used in the construction industry, where it is used to create structures and products.
9. It is also used in the manufacturing industry, where it is used to create products for specific applications.
10. It is also used in the pharmaceutical industry, where it is used to create drugs and other medical products.

4. Engineering Chemistry by K. Sesha Maheswaramma, Pearson

1. Engineering chemistry is a branch of chemistry that applies the principles of chemistry to the design, development, and production of engineering materials.
2. The field is concerned with the study of the structure, properties, and reactivity of materials, and the development of methods to manufacture and use them.
3. Engineering chemistry involves the study of the chemical processes involved in the production, processing, and use of materials.
4. It includes the study of the properties of materials, such as their strength, hardness, electrical and thermal conductivity, and optical properties.
5. It also includes the study of the chemical processes used in the production and processing of materials, such as oxidation, corrosion, and electrochemical processes.
6. Engineering chemistry also involves the development of new materials, such as polymers, composites, and nanomaterials, and the study of their properties and applications.

5. Engineering Chemistry by OG Palanna, Mc Graw Hill Education, New Delhi

1. Engineering Chemistry is a branch of chemistry that deals with the study of the principles and applications of chemical engineering.
2. It covers topics such as thermodynamics, kinetics, mass transfer, process design, and process control.
3. It also covers topics related to the production of chemicals, materials, and energy from raw materials.
4. It is used in the design and operation of chemical plants, as well as in the development of new products and processes.
5. It is also used in the development of new materials and the optimization of existing processes.
6. Engineering Chemistry is used in the design and operation of industrial processes, such as the production of pharmaceuticals, petrochemicals, and other chemicals.
7. It is also used in the development of new materials and the optimization of existing processes.
8. It is also used in the development of new products and processes.
9. Engineering Chemistry is an important part of chemical engineering, and is essential for the development of new technologies and products.

6. Engineering Chemistry by Shashi Chawala, Dhanpat Rai Publishing Comp, New Delhi

1. Engineering Chemistry is a subject focused on the application of chemical principles to engineering problems.
2. It covers topics such as thermodynamics, electrochemistry, corrosion, water treatment, and environmental chemistry.
3. It also covers topics such as the properties of materials, their production, and their uses in engineering.
4. The book covers the basic principles of engineering chemistry and their applications in engineering systems.
5. It also provides an in-depth understanding of the various chemical processes involved in engineering systems.
6. It also provides an insight into the various safety and environmental issues related to engineering chemistry.
7. The book provides an overview of the various types of engineering materials and their properties, and how they are used in different engineering systems.
8. It also covers the various methods of testing and analyzing engineering materials.
9. The book provides a comprehensive overview of the various chemical processes involved in engineering systems, and how they are used in different engineering applications.

7. University Chemistry by BH Mahan

1. Chemistry is the scientific study of matter and its properties, structure, composition, and behavior.
2. Chemical reactions involve the rearrangement of atoms to form new substances.
3. The laws of thermodynamics govern the energetics of chemical reactions.
4. Chemical bonds are the forces that hold atoms together in molecules and crystals.
5. Chemical equilibrium is the state in which there is no net change in the concentrations of reactants and products.
6. Kinetics is the study of the rates of chemical reactions and the factors that affect them.
7. Organic chemistry is the study of compounds that contain carbon and hydrogen.
8. Analytical chemistry is the study of the composition of substances.
9. Physical chemistry is the study of the physical properties of matter and the interactions between matter and energy.
10. Inorganic chemistry is the study of the properties and reactions of inorganic compounds.

8. University Chemistry by CNR Rao

- Chemistry is the study of matter, its properties, and how it interacts with other matter and energy.
- CNR Rao is an Indian chemist and a recipient of the Bharat Ratna, the highest civilian award in India.
- University Chemistry is a branch of chemistry that focuses on topics related to the study of matter, its properties, and its reactions with other substances.
- This branch of chemistry includes topics such as atomic and molecular structure, chemical bonding, thermodynamics, kinetics, electrochemistry, and spectroscopy.
- It also covers topics such as organic chemistry, inorganic chemistry, biochemistry, and physical chemistry.
- The study of university chemistry is essential for students who wish to pursue a career in the chemical sciences.